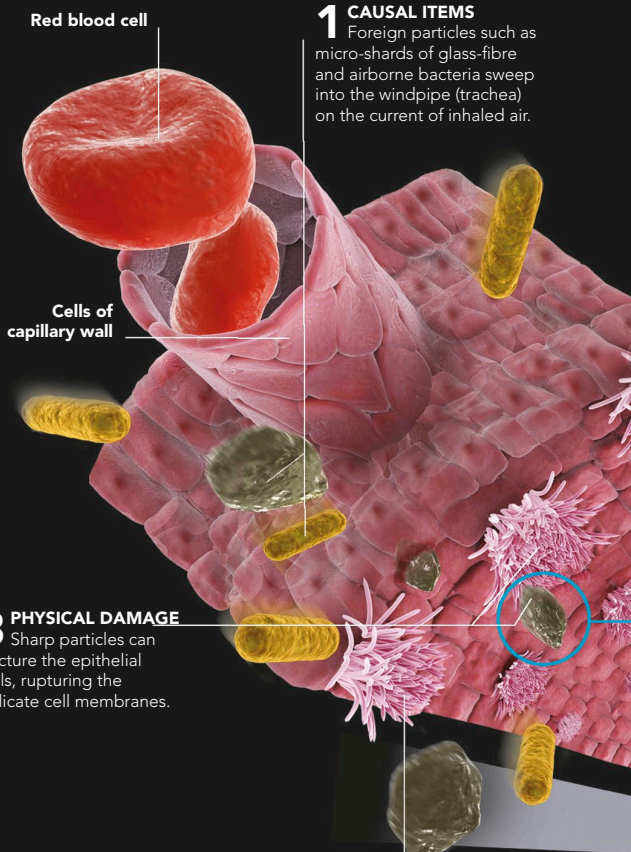
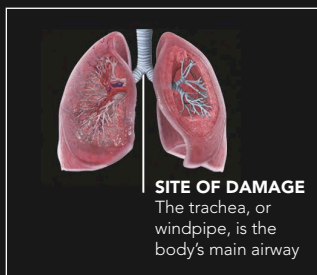


INFLAMMATORY RESPONSE

INFLAMMATION IS THE BODY'S RAPID, GENERAL RESPONSE TO ANY KIND OF INSULT OR INJURY, SUCH AS FROM PHYSICAL WOUNDS AND FOREIGN OBJECTS, INFECTING ORGANISMS, CHEMICAL TOXINS, HEAT, OR RADIATION.

The inflammatory response is a non-specific reaction that passes through defined phases and involves various types of white blood cell and defensive chemicals. The four cardinal signs are redness, swelling, heat, and pain. The process acts to attack, break down, and remove invading material, to dispose of the body's damaged cells and tissues, and to initiate healing.



CAUSE OF INFLAMMATION

The respiratory system is under constant threat from tiny inhaled particles of dust and debris and attack by infecting microbes. Here, the lining (epithelium) of the windpipe (trachea) mounts an inflammatory response to dust and bacteria. In reality, this process usually occurs alongside the specific immune response (see p.196), which targets individual foreign substances.